

RADHIKA AMAR DESAI

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. (CSE)	VIT, Chennai	9.24/10	2022-2026
Senior Secondary Certificate	CBSE Board	97.16%	2022

WORK EXPERIENCE

- Airbus** Jan'26 - Present
Computer Vision Intern Bangalore, Karnataka
 - Conducting research on coreset selection for data-efficient training of deep learning models, with emphasis on scalability and representation quality
 - Benchmarking coreset selection and data pruning methods across standard vision datasets to analyze performance, generalization, and computational trade-offs
 - Developing a novel embedding-space based coreset selection framework, currently under evaluation on benchmark datasets
 - Designed and deployed an object-level image retrieval Proof of Concept using AWS Cloud Services
 - Exploring image reconstruction and representation inversion from foundation model embeddings
- Stryker Global Technology Center R&D** Feb'25 - Jan'26
Research Associate Bangalore, Karnataka (Hybrid mode)
 - Contributing to a Proof of Concept (POC) in the medical imaging domain by integrating computer vision and deep learning approaches for clinical workflows.
 - Designed and implemented deep learning pipelines in PyTorch for tasks including image segmentation, image registration, temporal consistency, and inpainting, ensuring robustness and adherence to clinical data standards.
 - Built automated data preparation pipelines for large-scale medical imaging datasets, including preprocessing, augmentation, and annotation handling, leveraging Azure ML for reproducibility and scalability
 - Conducted model training, hyperparameter tuning, and experimentation on Azure ML, tracking performance metrics and optimizing for accuracy, efficiency, and deployment-readiness.
 - Collaborating with cross-functional teams to document findings and present insights for future clinical validation and productization.
 - Participating in labeling medical imaging datasets while coordinating and reviewing data annotation work, ensuring quality and consistency; provided guidance and feedback to fellow interns to maintain high annotation standards.
- Schneider Electric** May'24 - Jul'24
Project Trainee Doha, Qatar
 - Automated data preprocessing workflows for industrial control system logs (Foxboro controllers), improving efficiency and reducing manual effort.
 - Developed automation tools to test communication between Triconex and Foxboro systems, enhancing reliability in Emergency Shutdown Systems (ESD).
 - Created multiple Visual Basic automations for repetitive Excel tasks related to supply chain and internal reporting, improving efficiency for project teams.
 - Assisted in Factory Acceptance Testing (FAT) for large-scale industrial systems, gaining experience with audit-like validation processes.

RESEARCH PROJECTS

- Discriminative Span** Jan'25-Present
Research paper under preparation for submission to top-tier conferences on synthetic data
 - Developing a novel framework to analyze difference vectors in latent embedding spaces, quantifying the impact of synthetic data on downstream model performance.
 - Designing and implementing a modular pipeline integrating feature extraction (ResNet/EfficientNet/MobileNet), statistical analysis, and supervised classification, with ongoing ablation studies and extensions to domain adaptation and data selection.
 - Conducting extensive experiments on benchmark datasets (e.g., CycleGAN), evaluating alignment quality, classifier separability, and generalization, achieving up to ~96% test accuracy under high-alignment conditions
 - Paper: [Discriminative Span as a Predictor of Synthetic Data Utility via Classifier Reconstruction](#)
- Synthetic Data vs Generalization (CXr Study)** Jan'25-Present
Research paper under preparation for submission to top-tier conferences on synthetic data
 - Investigated the limitations of synthetic data in medical imaging, showing that models trained solely on synthetic CXr data fail to generalize despite strong performance on standard distributional metrics.
 - Developed a pseudo-paired synthetic data generation pipeline leveraging text-conditioned image editing for controlled pathology injection.

- Developed a task-aware evaluation metric to measure how well synthetic data represents real data by modifying the k-center greedy objective and incorporating uncertainty-based weighting.
- **Detection of Fraudulent Transactions In Ethereum** Oct'24-Apr'25
Research paper accepted into the International Conference On AI-Powered Security For Metaverse.
 - Developed an anomaly detection pipeline for fraudulent financial transactions using Python and deep learning, achieving 99.9% accuracy.
 - Designed data preprocessing and feature engineering workflows for large-scale financial datasets.
 - Repository: [Ethereum_Fraud_Transaction_Detection](#)
- **Automatic PCOS Detection System using Ultrasound Images of Ovaries** Jun'24-Jul'25
Research paper accepted into the International Conference On Communication And Smart Devices
 - Built a data preprocessing and anomaly detection pipeline for ultrasound datasets using Python and OpenCV.
 - Developed a novel segmentation technique to standardize noisy medical images.
 - Applied machine learning models (scikit-learn, ML libraries) achieving 99% accuracy in classification.
 - Paper: [Automatic_PCOS_Detection_Using_Ultrasound_images.git](#)

TECHNICAL SKILLS

- **Programming & Databases:** Python, SQL, C, C++, Visual Basic | **Data Analytics & Visualization:** Power BI, Excel, OpenCV | **Embedded:** Arduino | **Tools:** TriStation, FoxView | **Libraries:** PyTorch, OpenCV | **Cloud:** AWS, Azure ML | **Machine Learning & Others:** Data Preprocessing, Anomaly Detection, Image Segmentation, Automation Workflows

REFERRALS

- **VIT Research Faculties:** chanthini.baskar@vit.ac.in, modigari.narendra@vit.ac.in
- **Schneider Electric Project Manager:** ritwik.bhattacharjee@se.com
- **Stryker Senior Manager:** venkatachalam.vr@stryker.com